

# Fault Prognosis for Power Electronics Systems Using Adaptive Parameter Identification

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**Abstract**—This paper presents the design, implementation, and experimental validation of a method for fault prognosis for power electronics systems using an adaptive parameter identification approach. The adaptive parameter identifier uses a generalized gradient descent algorithm to compute real-time estimates of system parameters (e.g., capacitance, inductance, parasitic resistance) in arbitrary switching power electronics systems. These estimates can be used to monitor the overall health of a power electronics system and to predict when faults are more likely to occur. Moreover, the estimates can be used to tune control loops that rely on the system parameter values. The parameter identification algorithm is general in that it can be applied to a broad class of systems based on switching power converters. We present a real-time experimental validation of the proposed fault prognosis method on a 3 kW solar photovoltaic interleaved boost dc–dc converter system for tracking changes in passive component values. The proposed fault prognosis method enables a flexible and scalable solution for condition monitoring and fault prediction in power electronics systems.

**Index Terms**—Adaptive estimation, dc–dc power conversion, fault diagnosis, parameter estimation, power electronics.

## I. INTRODUCTION

MANY MISSION-CRITICAL power electronics systems, including renewable energy integration, data center power delivery, and motor drives applications, require high reliability and availability of service [1], [2]. In many of these scenarios, techniques for fault prognosis are commonly employed, that is, methods for actively monitoring the system condition and predicting when failures or faults will occur. A central technology that enables fault prognosis is parameter identification,

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TABLE I  
TYPICAL FAULT MODES FOR CAPACITORS (ELECTROLYTIC, CERAMIC, AND FILM) AND INDUCTORS (AIR INDUCTORS, FERRITES, AND IRON-CORE) IN POWER ELECTRONICS SYSTEMS [4], [7]

| Component  | Fault modes        | Fault type | Parameter drift direction             |
|------------|--------------------|------------|---------------------------------------|
| Capacitors | Open circuit (OC)  | Hard       | $C = 0$ , $ESR = \infty$              |
|            | Short circuit (SC) | Hard       | $C = \infty$ , $ESR = 0$              |
|            | Wear out           | Soft       | $C = \downarrow$ , $ESR = \uparrow$   |
| Inductors  | Open circuit       | Hard       | $L = 0$ , $ESR = \infty$              |
|            | Full-winding SC    | Hard       | $L = 0$ , $ESR = 0$                   |
|            | Inter-turn SC      | Soft       | $L = \downarrow$ , $ESR = \downarrow$ |
|            | Core to winding SC | Soft       | $L = \downarrow$ , $ESR = \uparrow$   |
|            | Wear out           | Soft       | $L = \downarrow$ , $ESR = \uparrow$   |

or identifying the values of system parameters in a real-time and online manner. By tracking the values of important system parameters in real time, operators can actively monitor the overall health of a system and anticipate when maintenance or repairs will be needed. Moreover, fault prognosis can be achieved by monitoring if estimated parameter values are above or below an accepted tolerance range.

The failure modes and mechanisms for power electronics systems have been widely investigated, for instance in [3]–[7]. Passive components, such as capacitors and inductors, are a key failure point. Table I provides an overview of the common failure modes of passive components in a power electronics systems and the effect that these failures have on the resulting parameter value and equivalent series resistance (ESR). The reasons for these failures vary widely, and include manufacturing defects, harsh environmental conditions (e.g., temperature and humidity), aging, high voltage stress, insulation failures, inter-connection failures, mechanical wear, vibrations and shocks [4], [6]–[8]. Moreover, the effect of the failures can be classified as either “hard” or “soft” faults. A hard fault is one that causes a sudden and catastrophic effect in the system (e.g., a short circuit), whereas a soft fault is one that causes a gradual effect or degradation in the system, generally related to lifetime wear or aging.

Parameter identification has been investigated previously in the context of power electronics systems. One salient application for parameter identification in power electronics systems has been for estimating the capacitance or ESR of a dc-link capacitor [9]–[14]. In many converters, the dc-link capacitor, particularly electrolytic capacitors, is one of the primary points of failure in the converter. Actively monitoring the capacitance

or ESR of the capacitor enables the detection and prediction of when these failures will occur.

In general, most of the algorithms for parameter identification compare measurements from the physical system (e.g., voltage, current, temperature) with a model structure. This model structure could be a black-box (e.g., neural networks [15]–[18]) or based on a physical model (e.g., Kalman filter, state observers [10], [19]–[24]).

These existing approaches in literature, however, have certain characteristics that limit the effectiveness and applicability of the methods. Some of these limitations are as follows:

- 1) high computational requirements (e.g., requires an external PC or graphical processing unit);
- 2) custom analog implementations;
- 3) techniques based on heuristic analysis that are applicable only to a specific switching converter;
- 4) injection of external voltage or current signals;
- 5) need for extensive data sets and training.

In this paper, we present a technique for fault prognosis for power electronics systems using an adaptive parameter identifier approach. The fundamental algorithm (first proposed in [25]) uses a switched linear model of a switching power converter and a generalized gradient descent algorithm to dynamically track the values of passive components, such as capacitors and inductors, in a power electronics converter. The algorithm is digitally implemented in real time on the same embedded processor as the control system, and is used to monitor when passive component parameter values are above or below a predefined tolerance range, which would indicate a fault scenario.

The design advantages of the proposed method compared to the state of the art is threefold. First, the modeling and identification algorithm is general in that it is (in theory) applicable to any switching power converter which can be modeled as a switched linear system. Thus, although the focus of this study is on dc–dc converters for solar photovoltaic (PV) applications, the technique can also be applied in the context of motor drive applications, cable integrity monitoring, among others. Second, the real-time algorithm is implemented digitally, as opposed to techniques that require custom analog implementations for each converter or estimated parameter. Moreover, the algorithm has low computational overhead, which enables it to be implemented on the same computing platform as the control system. Third, the proposed approach requires no additional sensors or computing devices (aside from those already used for closed-loop control purposes), or injection of external signals into the system.

The remainder of this paper is outlined as follows. Section II presents the modeling approach and algorithm design of the proposed adaptive parameter identifier. Section III presents a simulation of the identifier algorithm for an interleaved boost dc–dc converter. The simulation tests a variety of operating points and parameter perturbations, and provides a simulation analysis of the robustness of the algorithm to measurement noise and converter nonidealities. Section IV describes the hardware implementation and experimental testbed. Section V presents experimental results which validate the real-time implementation of the algorithm on a 3 kW dc–dc converter. Section VI concludes this paper.

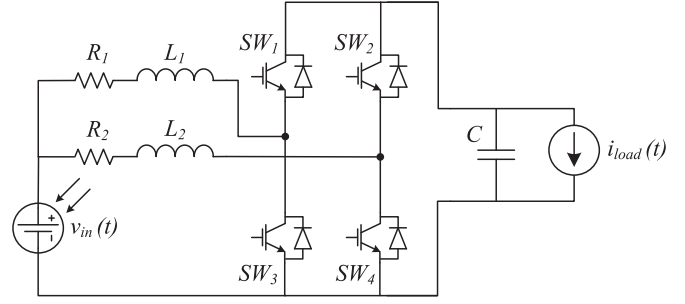


Fig. 1. Circuit topology of an interleaved boost dc–dc converter.

## II. ADAPTIVE PARAMETER IDENTIFIER DESIGN

In this section, we present the derivation of the proposed adaptive parameter identifier algorithm. We first present the modeling approach used to describe the nominal dynamics of a converter. Next, we use this model to analyze the dynamics of a converter when parameters are perturbed from their nominal values. Finally, we present a gradient descent algorithm that is used to track perturbed parameter values and to reconstruct estimates of the actual parameter values.

### A. Nominal Converter Model and State Estimator

Many power electronics systems consist of linear passive elements (e.g., capacitors, inductors, resistors) and switching devices (e.g., MOSFETs, IGBTs, diodes). Such systems can be modeled as a switched linear system, that is

$$\dot{\mathbf{x}}(t) = \mathbf{A}_{\sigma(t)} \mathbf{x}(t) + \mathbf{B}_{\sigma(t)} \mathbf{u}(t) \quad (1)$$

$$\mathbf{y}(t) = \mathbf{C} \mathbf{x}(t) \quad (2)$$

where  $\mathbf{A}_{\sigma(t)}$  and  $\mathbf{B}_{\sigma(t)}$  are the collection of linear state space models, and  $\sigma(t)$  is the continuous-time switching signal that indicates the active mode.

We can build an open-loop switched linear state estimator of the system from (1) as follows:

$$\dot{\mathbf{z}}(t) = \mathbf{A}_{\sigma(t)} \mathbf{z}(t) + \mathbf{B}_{\sigma(t)} \mathbf{u}(t) \quad (3)$$

where  $\mathbf{z}(t)$  is an estimate of the state vector  $\mathbf{x}(t)$  from (1) and  $\mathbf{A}_{\sigma(t)}$  and  $\mathbf{B}_{\sigma(t)}$  are obtained by solving Kirchhoff's circuit laws for the ideal circuit topology.

**Example 1:** Consider the interleaved boost converter topology in Fig. 1. In the continuous conduction mode of operation, the mode of the system is determined explicitly by the PWM applied to  $SW_1$  and  $SW_2$ . We can construct an open-loop state estimator of the system as follows:

$$\mathbf{z}(t) = \begin{bmatrix} \hat{i}_{L_1}(t) \\ \hat{i}_{L_2}(t) \\ \hat{v}_C(t) \end{bmatrix}, \mathbf{u}(t) = \begin{bmatrix} v_{in}(t) \\ i_{load}(t) \end{bmatrix}, \mathbf{y}(t) = \begin{bmatrix} i_{in}(t) \\ v_{out}(t) \end{bmatrix}$$

TABLE II  
POSSIBLE OPEN/CLOSE SWITCH POSITIONS FOR THE INTERLEAVED BOOST  
CONVERTER IN CONTINUOUS CONDUCTION MODE

| $\sigma(t)$ | 1 | 2 | 3 | 4 |
|-------------|---|---|---|---|
| $s_1$       | 0 | 1 | 0 | 1 |
| $s_2$       | 0 | 0 | 1 | 1 |
| $s_3$       | 1 | 0 | 1 | 0 |
| $s_4$       | 1 | 1 | 0 | 0 |

$$\mathbf{A}_{\sigma(t)} = \begin{bmatrix} -\frac{R_1}{L_1} & 0 & -\frac{s_1}{L_1} \\ 0 & -\frac{R_2}{L_2} & -\frac{s_2}{L_2} \\ \frac{s_1}{C} & \frac{s_2}{C} & 0 \end{bmatrix}, \mathbf{B}_{\sigma(t)} = \begin{bmatrix} \frac{1}{L_1} & 0 \\ \frac{1}{L_2} & 0 \\ 0 & -\frac{1}{C} \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The possible values for the switching signal  $\sigma(t)$  are given in Table II, where  $s_k = 0$  indicates switch  $\text{SW}_k$  is open, and  $s_k = 1$  indicates switch  $\text{SW}_k$  is closed.

#### B. Converter Model and Dynamics With Unknown Parameters

Consider the case when  $\mathbf{A}_{\sigma(t)}$  and  $\mathbf{B}_{\sigma(t)}$  are not precisely known. Let  $\delta\mathbf{A}_{\sigma(t)} \triangleq \hat{\mathbf{A}}_{\sigma(t)} - \mathbf{A}_{\sigma(t)}$ , where  $\hat{\mathbf{A}}_{\sigma(t)}$  is an estimate of  $\mathbf{A}_{\sigma(t)}$ , and  $\delta\mathbf{B}_{\sigma(t)} \triangleq \hat{\mathbf{B}}_{\sigma(t)} - \mathbf{B}_{\sigma(t)}$ , where  $\hat{\mathbf{B}}_{\sigma(t)}$  is an estimate of  $\mathbf{B}_{\sigma(t)}$ . The open-loop state estimator is now given by

$$\dot{\mathbf{z}}(t) = \hat{\mathbf{A}}_{\sigma(t)}\mathbf{z}(t) + \hat{\mathbf{B}}_{\sigma(t)}\mathbf{u}(t). \quad (4)$$

The dynamics of the error  $\mathbf{e}(t) = \mathbf{z}(t) - \mathbf{x}(t)$  are governed by

$$\dot{\mathbf{e}}(t) = \mathbf{A}_{\sigma(t)}\mathbf{e}(t) + \delta\mathbf{A}_{\sigma(t)}\mathbf{z}(t) + \delta\mathbf{B}_{\sigma(t)}\mathbf{u}(t). \quad (5)$$

Let  $\theta^*$  be a vector containing the actual value of the unknown parameters of interest, and  $\theta(t)$  be an estimate of  $\theta^*$ . In general, a switched linear state space model can be parameterized as follows:

$$\mathbf{A}_{\sigma(t)}\mathbf{x}(t) + \mathbf{B}_{\sigma(t)}\mathbf{u}(t) = \mathbf{W}_{\sigma(t)}(\mathbf{x}, \mathbf{u})\theta^* \quad (6)$$

where  $\mathbf{W}(\mathbf{x}, \mathbf{u})$  is a time-varying matrix that depends on the state and inputs.

We define  $\phi(t) \triangleq \theta(t) - \theta^*$  as the difference between the estimate and the actual value of the unknown parameters. Using this, we can rewrite (5) as follows:

$$\dot{\mathbf{e}}(t) = \mathbf{A}_{\sigma(t)}\mathbf{e}(t) + \mathbf{W}_{\sigma(t)}(\mathbf{z}, \mathbf{u})\phi(t). \quad (7)$$

We can solve for the time domain evolution of  $\dot{\mathbf{e}}(t)$  as follows:

$$\mathbf{e}(t) = \Phi(t_0, t)\mathbf{e}(t_0) + \phi \int_{t_0}^t \Phi(t, \tau)\mathbf{W}_{\sigma(t)}(\mathbf{z}, \mathbf{u})d\tau. \quad (8)$$

Note that we assume the term  $\phi(t)$  evolves slowly (essentially constant) compared to the state dynamics such that they appear

essentially constant; this allows us to bring the term outside of the integral. The corresponding error in the output  $\gamma(t) \triangleq \mathbf{C}(\mathbf{z}(t) - \mathbf{x}(t))$  implies that

$$\gamma(t) = \mathbf{C} \left( \Phi(t_0, t)\mathbf{e}(t_0) + \phi \int_{t_0}^t \Phi(t, \tau)\mathbf{W}_{\sigma(t)}(\mathbf{z}, \mathbf{u})d\tau \right). \quad (9)$$

Finally, the term  $\Phi(t_0, t)\mathbf{e}(t_0)$  asymptotically decays to zero, so the evolution of  $\gamma(t)$  is as follows:

$$\gamma(t) = \mathbf{H}(t)\phi \quad (10)$$

where  $\mathbf{H}(t) = \mathbf{C} \int_{t_0}^t \Phi(t, \tau)\mathbf{W}_{\sigma(t)}(\mathbf{z}, \mathbf{u})d\tau$ .

**Example 2:** Again, consider the interleaved boost converter topology shown in Fig. 1. The unknown parameters to be estimated are  $L_1$ ,  $L_2$ , and  $C$ . The resulting expressions for  $\mathbf{z}(t)$ ,  $\theta(t)$ , and  $\mathbf{W}_{\sigma(t)}(\mathbf{z}, \mathbf{u})$  are as follows:

$$\mathbf{z}(t) = \begin{bmatrix} \hat{i}_{L_1}(t) \\ \hat{i}_{L_2}(t) \\ \hat{v}_C(t) \end{bmatrix}, \theta(t) = \begin{bmatrix} \frac{1}{L_1(t)} \\ \frac{1}{L_2(t)} \\ \frac{1}{C(t)} \end{bmatrix}$$

$$\mathbf{W}_{\sigma(t)}(\mathbf{z}, \mathbf{u}) = \text{diag} \begin{pmatrix} -R_1\hat{i}_{L_1}(t) - s_1\hat{v}_C(t) - v_{in}(t) \\ -R_2\hat{i}_{L_2}(t) - s_2\hat{v}_C(t) - v_{in}(t) \\ s_1\hat{i}_{L_1}(t) + s_2\hat{i}_{L_2}(t) - i_{load}(t) \end{pmatrix}.$$

#### C. Gradient Descent Algorithm Design and Dynamics

The objective of the adaptive parameter identifier is to perturb the parameter estimate vector  $\theta(t)$  as to drive the measured output error  $\gamma(t)$  to zero. In this way, we expect  $\theta(t)$  to converge to the actual parameter values  $\theta^*$ .

There are a number of algorithms available for adaptive parameter identification [26]. For purposes of simplicity in design and implementation, we propose the use of a generalized gradient descent algorithm [25], [26]. Noting that  $\dot{\theta}(t) = \dot{\phi}(t)$ , the structure of the gradient descent algorithm is as follows:

$$\dot{\theta}(t) = -\mathbf{G}\mathbf{H}^T(t)\gamma(t) \quad (11)$$

where  $\mathbf{G} \succ 0$  is a positive definite matrix gain which is used to control the rate of convergence of the error term  $\gamma(t)$  to zero. The structure of  $\mathbf{G}$  is generally of the form  $\text{diag}(\epsilon_1, \dots, \epsilon_n)$ , where  $n$  is the number of terms to be identified, and each  $\epsilon$  term is parameter update gain for the associated unknown parameter.

The matrix gain  $\mathbf{G}$ , in general, influences the rate of convergence and the stability of the gradient descent algorithm. The design of  $\mathbf{G}$  depends on a number of factors (see [25] for a mathematical analysis). Among these include the following:

- 1) *Persistency of excitation of  $\mathbf{H}(t)$ :* The gradient descent algorithm requires that  $\mathbf{H}(t)$  satisfies a persistency of excitation condition [26]. Practically, this implies that the input  $\mathbf{u}(t)$ , the input voltage, and load current in Examples 1 and 2, are varying with time sufficiently such that

the dynamics of the system make the parameter identification possible. This condition is easily satisfied in many practical applications, for instance, in a solar PV system where both the input voltage and load current depend on time-varying irradiation and temperature of the solar panel. Conversely, a purely dc steady-state system would not allow the parameters to converge, as there would not be enough information in the  $\mathbf{H}(t)$  signal.

- 2) *Number of measurements versus number of unknown parameters*: The number of measurements available directly influences the excitation of  $\mathbf{H}(t)$ , i.e., more measurements mean higher excitation. Likewise, as the number of unknown parameters increases, the amount of excitation in  $\mathbf{H}(t)$  required for the gradient descent algorithm to converge also increases. Thus, it is generally straightforward to achieve convergence when the number of unknown parameters is less than or equal to the number of available measurements, as will be shown in Sections III and V. Applying the gradient descent problem to the scenario with more unknown parameters than available measurements can be an ill-posed problem.
- 3) *Desired convergence rate*: Each  $\epsilon$  is chosen such that the dynamics of  $\dot{\theta}(t)$  evolve much slower than the state dynamics, in order to satisfy the assumption made in (8). Moreover, in the case of multiple unknown parameters,  $\epsilon$  can be chosen such that all parameters exhibit similar convergence rates so certain parameters do not converge much faster than others.

Finally, we require an estimate of  $\mathbf{H}(t)$  as follows:

$$\hat{\mathbf{H}}(t) = \mathbf{C} \int_{t_0}^t \hat{\Phi}(t, \tau) \mathbf{W}_{\sigma(t)}(\mathbf{z}, \mathbf{u}) d\tau \quad (12)$$

where  $\hat{\Phi}$  uses updated parameters from  $\dot{\theta}(t)$  to provide an estimate of the state transition matrix  $\Phi$ .

The complete adaptive parameter identifier is shown as follows:

$$\dot{\mathbf{z}}(t) = \hat{\mathbf{A}}_{\sigma(t)} \mathbf{z}(t) + \hat{\mathbf{B}}_{\sigma(t)} \mathbf{u}(t) \quad (13)$$

$$\gamma(t) = \mathbf{C} \mathbf{z}(t) - \mathbf{y}(t) \quad (14)$$

$$\frac{d}{dt} \hat{\mathbf{H}}(t) = \hat{\mathbf{A}}_{\sigma(t)} \hat{\mathbf{H}}(t) + \mathbf{W}(\mathbf{z}, \mathbf{u}) \quad (15)$$

$$\dot{\theta}(t) = -\mathbf{G} \mathbf{H}^T(t) \gamma(t). \quad (16)$$

### III. SIMULATION AND ANALYSIS

In this section, a variety of MATLAB simulations are used in order to validate the performance and robustness of the proposed adaptive parameter identifier for a number of test cases and operating conditions. The interleaved boost dc–dc converter of Fig. 1 is used as the device-under-test. Table III presents the converter parameters and operating point for the nominal simulation cases.

In order to implement the adaptive parameter identifier in a simulation environment, (13)–(16) are discretized using the Euler method, and executed in a simulation loop with a fixed 500 ns time step. We inject perturbations in the output capacitor

TABLE III  
SIMULATION AND EXPERIMENT PARAMETERS

| Component parameters                           |                           |
|------------------------------------------------|---------------------------|
| $R_{1,2}$                                      | 0.082 $\Omega$            |
| $L_{1,2}$ (nominal)                            | 5.0 mH                    |
| $C$ (nominal)                                  | 2.85 mF                   |
| Simulated parasitic parameters (Section III-D) |                           |
| $R_{\text{esr}}$                               | 38 m $\Omega$ $\pm$ 50%   |
| $R_{\text{on}}$                                | 1 m $\Omega$              |
| $R_{\text{par1},2}$                            | 100 $\Omega$              |
| $R_{\text{par3}}$                              | 5 $\Omega$                |
| $C_{\text{par1}}$                              | 1 pF                      |
| $C_{\text{par2}}$                              | 1 nF                      |
| $L_{\text{par1}}$                              | 5.00 nH                   |
| $L_{\text{par2}}$                              | 1 nH                      |
| Nominal operating point                        |                           |
| $v_{\text{in}}(t)$                             | 100 V                     |
| $v_{\text{out}}(t)$                            | 380 V                     |
| $i_{\text{load}}(t)$                           | 2.5 A + 1 A, 10 Hz ripple |
| Converter switching frequency                  | 10 kHz                    |
| Adaptive parameter identifier gains            |                           |
| $\epsilon_1, \epsilon_2$ ( $L_1(t), L_2(t)$ )  | $1 \times 10^6$           |
| $\epsilon_3$ ( $C(t)$ )                        | $3 \times 10^6$           |
| Embedded computing platform                    |                           |
| Device                                         | Zynq-7000 SoC XC7Z020     |
| Fixed computation time step                    | 500 ns                    |
| Voltage sensor bandwidth                       | 100 kHz                   |
| Current sensor bandwidth                       | 200 kHz                   |

and a series input inductor of the plant converter using additional shunt and series elements. For the purposes of fault monitoring, one can define a lower and upper bound of the acceptable range of parameter value. When the estimated parameter value leaves the acceptable range, a fault flag can be raised.

The output measurements, unknown parameter vector, and gain matrix from Examples 1 and 2 are used in the simulation, that is

$$\mathbf{y}(t) = \begin{bmatrix} i_{\text{in}}(t) \\ v_{\text{out}}(t) \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\boldsymbol{\theta}(t) = \begin{bmatrix} \frac{1}{L_1(t)} \\ \frac{1}{L_2(t)} \\ \frac{1}{C(t)} \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} \epsilon_1 & 0 & 0 \\ 0 & \epsilon_2 & 0 \\ 0 & 0 & \epsilon_3 \end{bmatrix}.$$

For simulation and experiments, we estimate either one or two parameters at a time. As discussed in Section II-C, since only two measurements are made in the simulations  $i_{\text{in}}(t)$  and  $v_{\text{out}}(t)$   $\mathbf{H}(t)$  would not have enough excitation and  $\boldsymbol{\theta}(t)$  would not converge if there are more than two unknown parameters. For these scenarios, the  $\epsilon_i$  associated with the  $i$ th desired parameter(s) to be estimated are set to the value shown in Table III, and the  $j$ th parameter(s) not being estimated has  $\epsilon_j = 0$ .

Moreover, in all simulation cases, a 1 A peak-to-peak ripple at 10 Hz is added to the 2 A load current  $i_{\text{load}}(t)$ . In addition



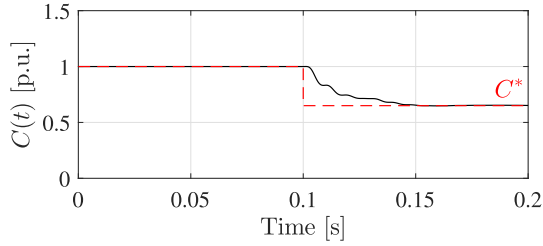


Fig. 2. Simulation of the adaptive parameter identifier tracking a step perturbation in capacitance  $C$  ( $t = 0.1$  s from  $C$  to  $0.65 C$ ).

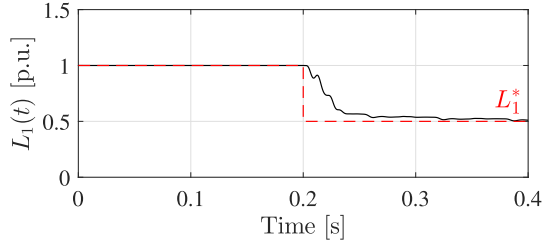


Fig. 3. Simulation of the adaptive parameter identifier tracking a step perturbation in inductance  $L_1$  ( $t = 0.2$  s from  $L_1$  to  $0.5 L_1$ ).

to emulating time-varying load dynamics, this ripple introduces the persistency of excitation to  $\mathbf{H}(t)$  that is necessary for the convergence of the error term  $\gamma(t)$  to zero. A similar perturbation could be added to the input voltage  $v_{\text{out}}(t)$  that would emulate the time-varying PV irradiation.

#### A. Step Perturbation in Single Parameter

One application of the proposed adaptive parameter identifier is the tracking of “hard” faults in passive components, that is drastic changes in the parameter value indicative of a catastrophic component failure. To simulate these types of faults, we introduced a step perturbation in the parameter value of passive components in the plant. As shown in Fig. 2, when a 35% step perturbation introduced in the capacitance  $C$ , the estimate  $C(t)$  converges to the correct value in around 50 ms. Similarly, when a 50% step perturbation introduced in the series inductance  $L_1$ , the estimate  $L_1(t)$  converges to the correct value in around 100 ms.

#### B. Ramp Perturbation in Single Parameter

Another application of the proposed adaptive parameter identifier is the tracking of slow varying perturbations in the parameter values of passive components. These types of slow varying changes occur on the time scale of hours to years, and represent a type of “soft” fault, that is, gradual changes indicative of lifetime wear or aging.

We simulated ramp perturbations in the model to emulate these types of faults. First, a ramp perturbation of  $-166.7 \mu\text{F/s}$  is introduced in the output capacitor  $C$ . This is a relatively fast perturbation meant to identify the upper limit of accuracy of the adaptive parameter identifier. Slower perturbations are tracked at equal or better accuracy. As shown in Fig. 4, the adaptive parameter identifier tracks the correct value of this

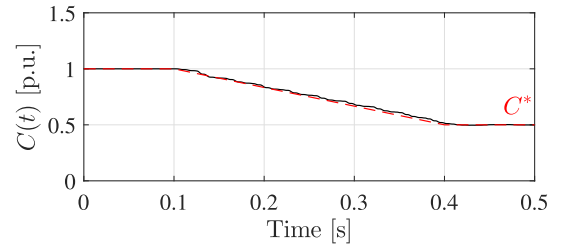


Fig. 4. Simulation of the adaptive parameter identifier tracking a ramp perturbation in capacitance  $C$  ( $-166.7 \mu\text{F/s}$  from  $C$  to  $0.5 C$ ).

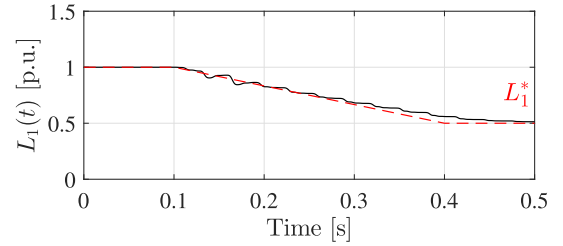


Fig. 5. Simulation of the adaptive parameter identifier tracking a ramp perturbation in inductance  $L_1$  ( $-16.7 \text{ mH/s}$  at  $t = 0.1$  s from  $L_1$  to  $0.5 L_1$ ).

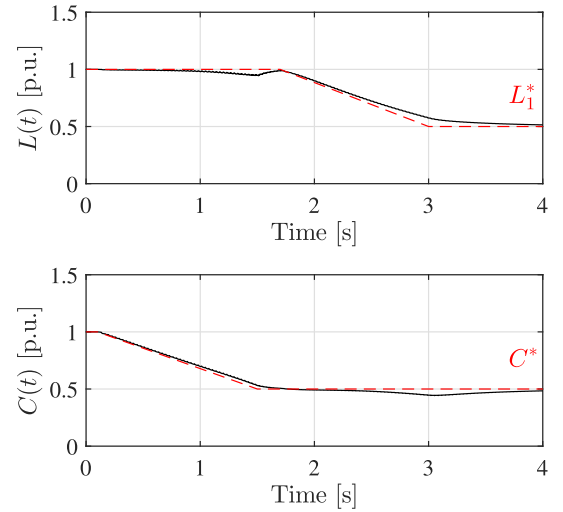


Fig. 6. Simulation of adaptive parameter identifier tracking simultaneous ramp perturbations in inductance  $L_1$  ( $-1.9 \text{ mH/s}$  at  $t = 1.75$  s) and capacitance ( $-35.7 \mu\text{F/s}$  from  $C$  to  $0.5 C$  at  $t = 0.2$  s).

perturbation with unnoticeable error. Similarly, we introduced a ramp perturbation of  $-16.7 \text{ mH/s}$  in a series input inductor  $L_1$ . As shown in Fig. 5, the adaptive parameter identifier again tracks the correct value of this perturbation.

#### C. Simultaneous Perturbation in Multiple Parameters

In general, more than one unknown parameter will be of interest in the system. From the analysis in Section II, we expect that the convergence rate of  $\theta(t)$  will be slower than if only one parameter is being estimated. This is due to the corresponding increase in the error  $e(t)$  and decrease in the excitation in  $\mathbf{H}(t)$ .

To test this scenario, we simulated a simultaneous ramp perturbation in both the output capacitor  $C$  and a series input in-

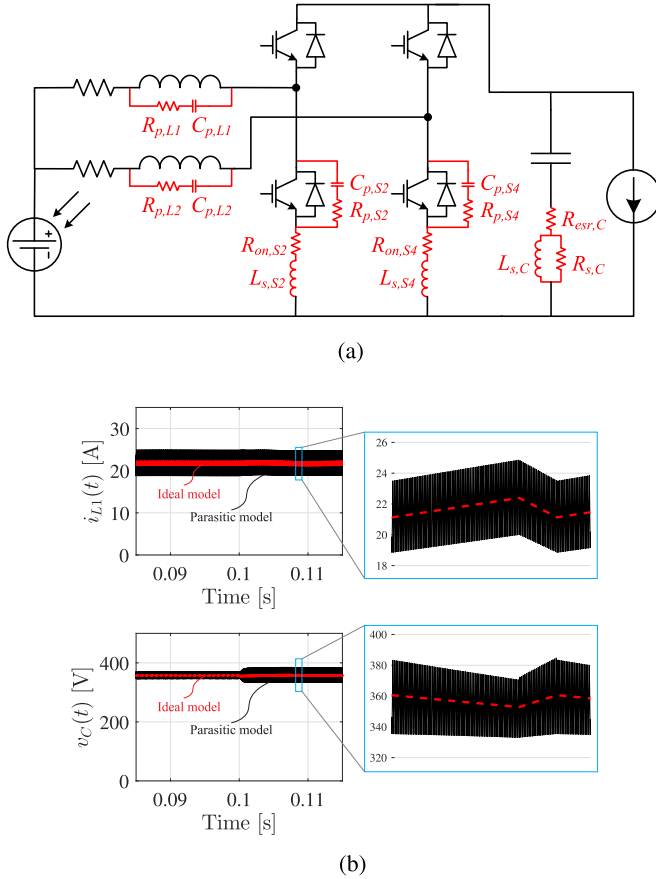


Fig. 7. Simulation of an interleaved boost dc-dc converter incorporating switch and passive component parasitics and nonidealities is used to validate the robustness of the proposed adaptive parameter identifier. (a) Circuit topology of the parasitic simulation model. (b) Steady-state dynamics of the ideal (from Fig. 1) and parasitic simulation model.

ductor  $L_1$ . The rate of change of these parameter values was set at  $-1.9$  mH/s and  $-35.7$   $\mu$ F/s, respectively. As shown in Fig. 4, the adaptive parameter identifier tracks the correct value of this perturbation with unnoticeable error. Slower perturbations are tracked at equal or better accuracy.

#### D. Effect of Measurement Noise and Circuit Nonidealities

Simulation was also used to test the robustness of the proposed adaptive parameter identifier to measurement noise and parameter and model nonidealities. The modified circuit model shown in Fig. 7(a) is used to test this scenario. Note that the adaptive parameter identifier still uses the “ideal” circuit model in Fig. 1, while the modified circuit model provides the measurement vector  $\mathbf{y}(t)$ . As shown, the modified circuit model incorporates switch and passive component parasitics, including the ESR of the output capacitance  $R_{esr,C}$ . The steady-state dynamics are shown in Fig. 7(b). As shown, when compared with the steady-state dynamics of the ideal model, the parasitic model introduces high-frequency current transients of approximately  $4$  A<sub>pk-pk</sub> in the series inductor currents  $i_L(t)$ , and voltage transients of approximately  $20$  V<sub>pk-pk</sub> in the output capacitor voltage  $v_C(t)$ .

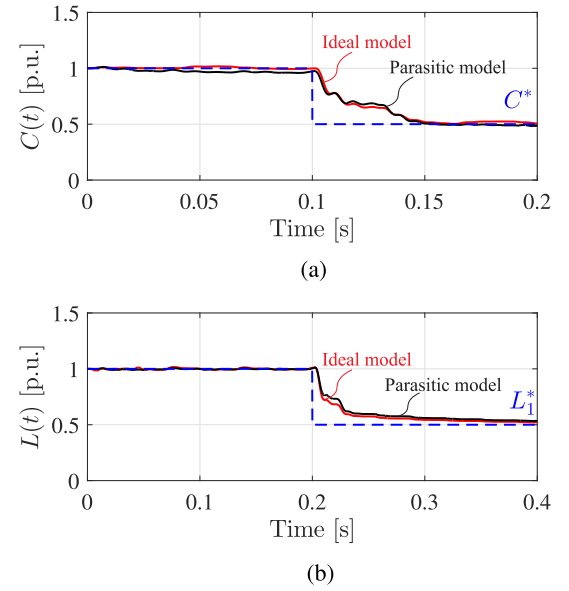


Fig. 8. Comparison of the adaptive parameter identifier convergence when applied to the ideal model (Fig. 1) versus parasitic model [Fig. 7(a)]. (a) Step perturbation at  $t = 0.1$  s from  $C$  to  $0.5C$ . (b) Step perturbation at  $t = 0.2$  s from  $L_1$  to  $0.5L_1$ .

We study the impact of the parameter and model nonidealities and parasitics on the convergence of the adaptive parameter identifier. The parameter step perturbation in Section III-A is used as a baseline to compare the ideal and parasitic models. As shown in Fig. 8, the parasitic model causes minimal difference in the dynamics of the adaptive parameter identifier for parameter step changes in both the output capacitor  $C$  and a series inductor  $L_1$ . This provides evidence that the algorithm is robust to the nonideal dynamics introduced by the parasitics.

A central reason for this is the design of the adaptive parameter identifier. In (8), we assumed that the  $\phi(t)$  evolves slowly (essentially constant) compared to the state dynamics. Dynamics introduced by parasitics or measurement noise would occur on a time scale faster than that of the state dynamics, and would be ignored by the parameter identification.

#### IV. HARDWARE IMPLEMENTATION AND EXPERIMENTAL TESTBED

In this section, we present the hardware implementation and experimental testbed for the proposed fault prognosis algorithm. The complete testbed is shown in Fig. 9. The complete specifications for the experimental testbed and computing platform are given in Table III.

##### A. Real-Time Digital Implementation

The fault prognosis algorithm is implemented in real time on the same computing device as the control system, in this case, a ZedBoard system-on-chip (SoC) device, a relatively low-cost development board for the Xilinx Zynq-7000 SoC XC7Z020. A central feature of this SoC is the integration programmable logic and a dual-core ARM Cortex-A9 processor. It is worth noting

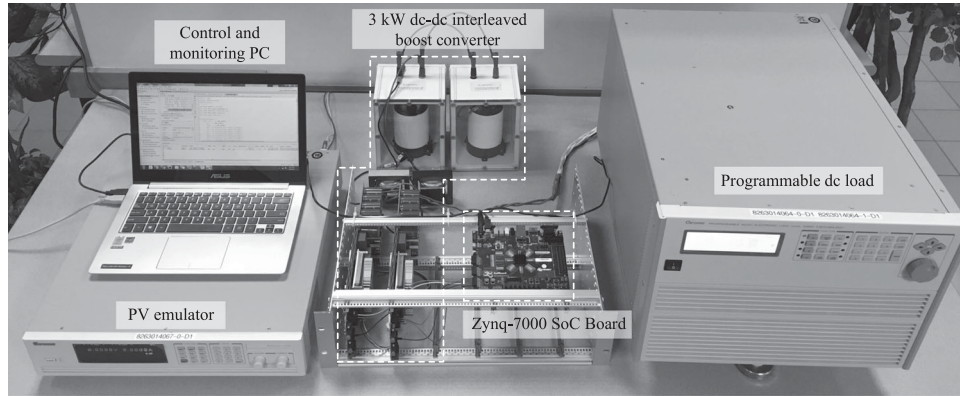


Fig. 9. Photograph of the experimental testbed, including the interleaved dc–dc converter and the Zynq-7000 SoC board.

that our implementation only utilized 11% of the programmable logic slices of the device. Moreover, the ARM core was not used in the design. Thus, the algorithm could be implemented on a much lower cost FPGA. In general, however, a programmable logic device is preferred for implementation (as opposed to a standard microcontroller) due to the tight constraints on latency and computation time imposed by the fault prognosis algorithm.

The adaptive parameter identifier in (13)–(16) is discretized and solved in realtime with a fixed time step (including input/output latency) of 500 ns. This time step, which is an order of magnitude faster than the switching frequency of the converter (10 kHz), is necessary to adequately model the switching dynamics of the converter.

Moreover, a highlight of the approach and implementation is that it maintains the flexibility to be reconfigured for different converters simply by changing the contents of the  $\hat{\mathbf{A}}_{\sigma(t)}$ ,  $\hat{\mathbf{B}}_{\sigma(t)}$ , and  $\mathbf{W}(\mathbf{z}, \mathbf{u})$  matrices. The selection of the  $\theta(t)$  vector indicates the parameters of interest to be estimated.

### B. Experimental Testbed

We experimentally validate the proposed fault prognosis method on a 3 kW interleaved dc–dc converter, as shown in Fig. 9. The input of the converter is connected to a programmable PV emulator, and the output is connected to a programmable dc load. Prior to running the experiments, the values of passive components are measured using an LCR meter. The setup is equipped such that the value of the output capacitor  $C$  and a series inductor  $L_1$  can be changed in real time by controlling solid-state switches that introduce series or parallel elements into the circuit.

## V. EXPERIMENTAL RESULTS

This section presents experimental results which validate the proposed fault prognosis algorithm under a variety of operating conditions.

We test step changes in the value of the output capacitor  $C$  and a series inductor  $L_1$ . Similar to the simulations performed in Section III-A, the hard fault instantaneously changes the value of the capacitor and the inductor, and we monitor the convergence

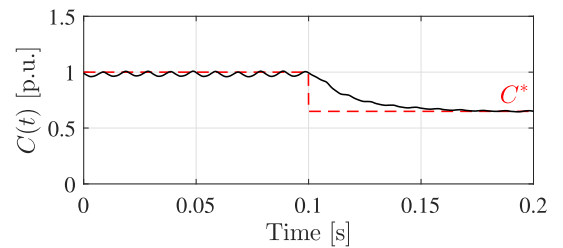


Fig. 10. Experiment of adaptive parameter identifier tracking a step perturbation in capacitance  $C$ . Step perturbation at  $t = 0.1$  s from  $C$  to  $0.65 C$ .

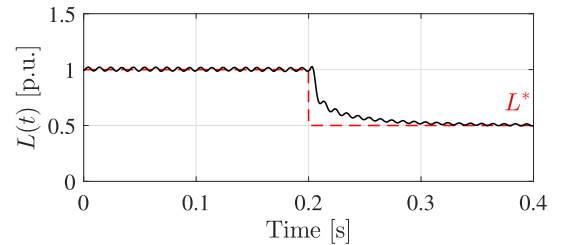


Fig. 11. Experiment of adaptive parameter identifier tracking a step perturbation in inductance  $L_1$ . Step perturbation at  $t = 0.2$  s from  $L$  to  $0.5 L$ .

of the adaptive parameter identifier to the new value. Moreover, we test the effect of changing the dynamics of the load current  $i_{\text{load}}(t)$  on the speed of convergence of  $\theta(t)$ .

For the following experiments, the same output measurements, unknown parameter vector, and gain matrix are used as in Section III. Relevant parameter values and operating points can be found in Table III.

### A. Step Perturbation in $C$ and $L_1$

Fig. 10 shows a fault that causes the output capacitance  $C$  to decrease from 2.85 to 1.85 mF, a decrease of 35%. As shown, the adaptive parameter identifier converges to the new value of  $C$  in approximately 50 ms. Similarly, Fig. 11 shows a fault that causes a series inductor  $L_1$  to decrease by 50%. Convergence to the new value of  $L_1$  takes approximately 100 ms. These values match the expected time of convergence from simulations in Section III-A.

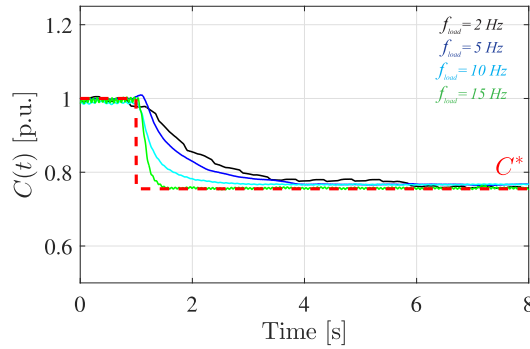


Fig. 12. Experiment of adaptive parameter identifier tracking step perturbations in capacitance  $C$  for a set of  $i_{load}(t)$  frequencies: 2, 5, 10, and 15 Hz. Step perturbation at  $t = 1$  s from  $C$  to  $0.65 C$ .

### B. Effect of Dynamic Operating Conditions

Next, we varied the frequency of the load current  $i_{load}(t)$  and observed its effect on the convergence of  $\theta(t)$ . We expect that a lower  $i_{load}(t)$  frequency will result in less excitation in  $\mathbf{H}(t)$ . Thus, with a constant matrix gain  $\mathbf{G}$ , the rate of convergence of  $\theta(t)$  will be slower. For higher  $i_{load}(t)$  frequencies, we expect the converse to be true.

We inject a step change in the output capacitance  $C$  from 2.85 to 1.85 mF for a set of  $i_{load}(t)$  frequencies: 2, 5, 10, and 15 Hz. Fig. 12 shows the results of this test. As shown, when the load is perturbed at 15 Hz, convergence to the new value of  $C$  takes approximately 50 ms. At slower perturbation frequencies (e.g., 2 Hz), convergence to the new value of  $C$  can take as long as 3 s. These results match our expectations, and also provide an empirical method for determining the rate of convergence of  $\theta(t)$  given a certain input vector  $\mathbf{u}(t)$ .

## VI. CONCLUSION

This paper presents a method for fault prognosis using an adaptive parameter identification approach. The salient advantages of the approach include application flexibility and the obviated need for additional sensors, computing devices, or injected signals. The proposed technique can be applied in domains, including motor drive applications (for stator or rotor resistance estimation), cable integrity monitoring, or other applications, where parameter or system identification is a useful technique for fault prognosis or prediction.

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